

SULIT

DC014

*Computer Science
Semester I
Session 2023/2024
2 hours*

DC014

**Sains Komputer
Semester I
Sesi 2023/2024
2 jam**

Nama	No. Matrik										Praktikum

(Sila isi maklumat dengan jelas dan lengkap)



KOLEJ MATRIKULASI LABUAN
LABUAN MATRICULATION COLLEGE

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

SAINS KOMPUTER
2 jam

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
No. Soalan	Markah	
	Pemeriksa	KP / KKP
1 (6 M)		
2 (16 M)		
3 (6 M)		
4 (4 M)		
5 (18 M)		
6 (20 M)		
Jumlah	/ 70	

Kertas soalan ini mengandungi **5** halaman bercetak
This question paper consist of 5 printed pages

INSTRUCTIONS TO CANDIDATE:

This question paper consists of 6 questions.

Answer **all** questions in the space provided.

The use of non-programmable scientific calculator is permitted.

ARAHAN KEPADA CALON:

Kertas soalan ini mengandungi 6 soalan.

*Jawab **semua** soalan di ruang yang disediakan.*

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

1. You are tasked with designing a website for a local business using multimedia elements to enhance user engagement.

a) Describe one (1) multimedia element that you would incorporate into the website.

[2 marks]

b) Explain the purpose of the multimedia element mentioned in 1(a) in the context of a business website.

[2 marks]

c) Give one (1) difference between three dimensional (3D) and four dimensional (4D) in terms of axis

[2 marks]

Three dimensional (3D)	Four dimensional (4D)

2. Imagine you are the project manager for a multimedia production tasked with creating an educational video series for a client. The client has requested a series of engaging videos to teach complex scientific concepts to high school students. Your team includes content creators, scriptwriters, animators, and voiceover artists. You have just started the project, and the client expects a detailed for whole phases.

a) (i) Briefly describe the importance of planning in pre-production phase

[2 marks]

(ii) Based on the description below, identify the Multimedia phases.

[4 marks]

Description	Phase
Fast & furious 9 movie, make shooting video based on the edited scripts.	
Flowcharts and storyboard are used as outline/blueprint for develop Web site project.	
Haziq & Aina movie are distributed in the form of compact discs or through websites.	
Regarding this statement "The first question we ask is "why" you want to develop a multimedia project".	

- b) Authoring tools are any software that can help teachers or students to create digital content. These tools are most commonly used to create e-learning content like online lessons, courses and videos.

(i)

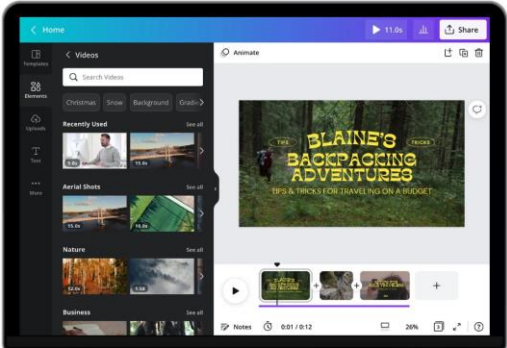


Figure 1

[4 marks]

Figure 1 shows a screenshot of a multimedia software authoring tool. State two (2) type authoring tools available that are provided by Canva and describe one (1) characteristic of each of the authoring tools mentioned.

	A	B
Type		
Characteristic		

c)



Figure 2

Figure 2 shows the screenshot of navigation used to obtain information from one related topic to another by using the logical flow of the interactive interface.

- (i) State the type of navigation based on Figure 2.

[1 mark]

- (ii) Explain the navigation that is mentioned above.

[2 marks]

- d) Multimedia lessons can be delivered in multiple ways, including stand-alone methods and can be also to place multimedia lessons on the World Wide Web as a part of an e-learning program.

(i) State one (1) example of delivering multimedia content through the World Wide Web.
[1 mark]

(ii) Explain one (1) disadvantage of medium delivery mentioned in 2(d)(i)
[2 marks]

3. A database is an organized collection of structured information, or data, typically stored electronically in a computer system

a) State two (2) approach to manage and store the data.
[2 marks]

Approach A :

Approach B :

b) Give two (2) differences between the approach mentioned above based on the terms given.
[6 marks]

	Approach A	Approach B
Data redundancy		
Storage		
Disadvantage		

4. A database object in a relational database is a data structure used to either store or reference data.

a) Based on the statement given, state the appropriate database object.
[4 marks]

	Scenario	Database Object
(i)	It consists of simple English-like statements that allow users to specify the data to display, print or store.	
(ii)	Allows to group and summarize data in chart form.	
(iii)	A screen that provides an area for entering or modifying data in the database.	
(iv)	Collection of records that identify a category of data, such as Students, Lecturers, or Academics	

5. In information technology, a network is defined as the connection of at least two computer systems. Each computer has access to the data of the other device and can share resources such as disk space, applications or peripheral devices

a) (i) Explain how a network is used in Multimedia Networking.

[2 marks]

(ii) Describe common uses of networking applications in the Global Positioning System (GPS).

[2 marks]

b) Figure 3 below shown the wired and wireless of transmission media.



Figure 3

(i) Name the picture for Type A and Type B.

[2 marks]

Type A	Type B

c) State the suitable type of wired transmission media based on usage below.

[4 marks]

	Usage	Wired transmission media
(i)	It is used in telephone lines to provide data and voice channels.	
(ii)	It is made up of thin strands of glass used to transmit information and data in the form of light.	
(iii)	It is used for broadband internet and cable television services, providing high-speed data transmission and reliable connectivity.	
(iv)	It is utilized for transmitting high-quality audio and video signals in professional broadcasting, home theatre systems, and security camera installations.	

d) Explain one (1) type of transmission media that carries signal through substances without physical material on a network.

[2 marks]

e) Figure 4 below shows a basic network diagram.

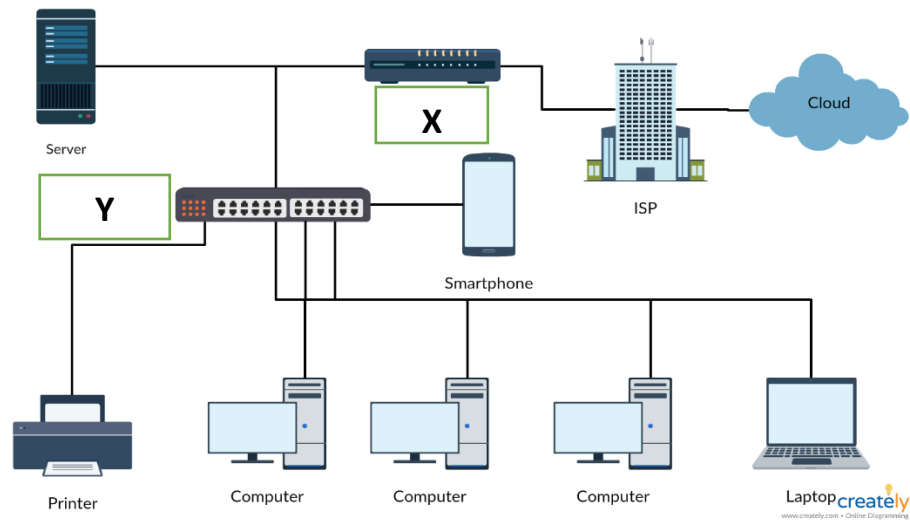


Figure 4

(i) Identify communication device X and Y.

[2 marks]

X : _____

Y : _____

(ii) Describe the function of communication device Y mentioned in question 5(e)(i).

[2 marks]

6) The Internet is a global network of billions of computers and other electronic devices. With the Internet, it's possible to access almost any information, and communicate with anyone else in the world.

a) (i) Explain the characteristics of two (2) types of search tools.

[4 marks]

Types of Search tools		
Characteristics		

- (ii) Give one (1) example for each search tools.

[2 marks]

- b) State suitable search strategy based on statement.

[4 marks]

	Statement	Search strategy
(i)	AND, OR and NOT are used to create a very broad or very narrow search	
(ii)	Type the keyword, "corporate social responsibility", the search engine will only bring back results that have all those words.	
(iii)	Use logic-based words that help search engines narrow down or broaden search results.	
(iv)	Limits the search to a particular search characteristic (or field), such as the page title, URL, page text, top-level domain, or Web site	

- c) The Internet makes it possible to quickly find information, communicate with people around the world, manage your finances, shop from home, listen to music, watch videos and much more.

[4 marks]

	Description	Type of web activities
(i)	Learning activities in which students interact with resources, or other students, or both, using the capabilities of the Internet or other computer-based communication networks.	
(ii)	Using internet-based sites to stay connected with friends, family, colleagues, or customers. It used for social communication purpose or a business purpose, or both through sites like Facebook, Twitter, Instagram, and Pinterest.	
(iii)	Customers access an online store to browse through and place orders for products or services via their own devices.	
(iv)	Downloading, playing, and enjoying the media such as games, music, and video from the Internet.	

- d) Nowadays, transactions and daily activities are being performed by online, there is the potential for many private information to be collected and distributed without the individual's knowledge or permission.

- i) Describe Web browsing privacy issues concerning the Internet.

[2 marks]

-
- ii) Emails have become such an integral part of our private and business lives that we consider them as intimate as having a conversation with a trusted friend.

State two (2) tips on how to protect your email privacy.

[2 marks]

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-
- e) Explain Internet of Thing (IoT) and state one (1) example of IoT use in plantation or agriculture.

[2 marks]

END OF QUESTIONS